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Evaluating Jogging Routes in Mass Models

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ABSTRACT

Designers and Urban Planners use various tools and metrics to help guide their decisions. Evaluating an urban environment based on distance to amenities (i.e., a source and destination) is only part of the movement experience of an occupant. In recreational movement such as walking, jogging, and running, people plan a path in which the start and end point is the same. The route taken and the experience along this route has implications for the satisfaction of the occupant. This paper introduces a straightforward method of generating all possible routes via two waypoints that meet a user-defined criteria for the total length of the route based on travel speed and time. Two new evaluative metrics for the paths are introduced: the percentage of the path that is in shade as well as the percentage that contains the Sun in the glare range of a person. As the latter aspect is a function of direction, a directed graph is used to factor in the directionality of the recreational route. Similarly, the truest location of the person is required for evaluating the impact of the built environment so the sidewalk, rather than road centerlines, are used.

Author Keywords

human factors, walkability, computation, recreation, evaluation, urban planning

ACM Classification Keywords

I.6 SIMULATION AND MODELING : [General]; J.5 ARTS AND HUMANITIES: [Architecture]; D.2.2 SOFTWARE ENGINEERING: Design Tools and Techniques—*Human Factors*; I.3.6 COMPUTER GRAPHICS: Methodology and Techniques—*Ergonomics*

1 INTRODUCTION

The goal of this paper is to provide an evaluation of the virtual environment to a designer that encompasses the availability of comfortable jogging routes. While existing tools evaluate human experiences based on visibility to other locations or distances to transportation, there is yet to be a focus on occupant physical and visual experience based on solar lighting. Unlike real-time path planners, the goal is not to find an optimal or most comfortable route, but rather evaluate an exhaustive set of routes formulated based on the environment (in particular, the sidewalk and crosswalk surfaces) given. Using these routes (e.g., (Fig. 1), various metrics can be calculated and provide a score— not of the route itself, but of the environment that led to those routes being created.

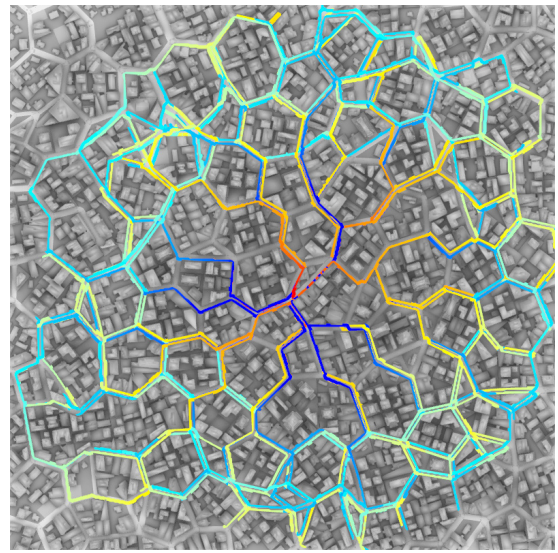


Figure 1: Overlay of 300 jogging paths generated. A color gradient is applied to each path corresponding to the start and end of the individual path color blue to red respectively.

When considering an urban environment for analysis related to the occupants, walkability is often at the forefront. The term walkability is a topic of research in numerous fields [1], yet even within urban design itself there are various meanings [9]. While specific terms and factors being analyzed vary, there is an underlying unification in goals that the built environment can influence both the behaviour and the health of the people within – albeit physical or mental.

Human subject studies are vital in the development of design process and thinking as they help to refine, and at times reject, the intuitive knowledge designers bring to the process. For example, a past study using infrared cameras and tracking of pedestrians throughout the trails in a city found an unexpected relationship that countered the urban design theory in which shorter street blocks induce higher levels of activity [12]. Research has shown the facilitation and increased recreational movement within an environment can be brought about by a variety of nature-based settings [15]. Consequently, it is feasible to consider the uncontrollable factors of jogging routes (i.e., the sun) when evaluating an environment, as additional interventions (e.g., creating a park or modifying the sidewalk surface condition) are suitable for increased activity. There is also an element of perceived vs. actual

attributes of an environment as Zuniga-Teran et al. (2019) argues; the perception of a neighborhood being pedestrian friendly is linked to visitation rates of greenspace [23].

Along with the risks (e.g., skin cancer), the benefit of sun exposure has been detailed in literature [16]. As the preferences and health benefits to every combination of sun and lighting conditions is a complex and still developing knowledge, this paper takes a computational approach to generating the possibilities and presenting the results, without determining the value of one over another. The proposal is then to develop analysis methodologies and techniques that find a large number of possible routes within a given constraint in order to analyze the environmental conditions one would experience along these routes.

With the existing literature emphasizing the importance of restoration and facilitation of the built environment for recreational mobility as a means towards a healthy community, the work presented here is positioned within the neighbourhood. The jogging routes calculated have a source and destination of the same location as the assumption placed is the person jogging, does so from their home, and returns back.

To avoid confusion with nuances in meaning, the terms; route, trail, and path, are used interchangeably to refer to an ordered list of locations a person travels along. The various terms are all used to stay consistent in terminology with the original authors of the literature review. In the computational perspective the term path is often used to describe an ordered list of nodes as it is derived from shortest-path algorithms, the underlying method used in this research. Recreational routes refers to walking, jogging, or running for leisure rather than an alternative mode of transportation. The emphasis on jogging is somewhat arbitrary for the methods in this research, but in terms of distance and speeds used in the results, it is the most similar type of activity.

1.1 Contribution

This paper focuses on route planning of closed cycles (Fig. 1) in a graph for evaluating the environment itself, rather than the path created. The metrics often used in route planning for filtering options for a user are stored as the layered costs for each route found to satisfy a given distance. Additionally, the metrics of route percentage that is in shade and in glare is demonstrated as a new addition to recreational routing evaluation. The evaluation system is written as a plugin to the Rhino 3D software in which a user can input their routing constraints and visualize aggregate or individualized data through Grasshopper.

The methods presented can benefit the urban design and planning field by evaluating and visualizing data related to the possible routes for recreational activities. This data can be used as an aid in decision making for defining areas of safer and pedestrian-centric intersections/crosswalks as well as unique (possibly more costly and with limited budgets) sidewalk materials and maintenance schedules. The methods for evaluating the paths are applicable to the evaluation of any route. However, the emphasis on jogging with route generation is due to the existing literature showing the circular path

attributes to be most applicable to this type of recreational path.

2 RELATED WORK

2.1 User Tracking

The impact of the built environment on a persons perception of their running route has been a topic of recent research, although it is still lacking. While there has been a longer history of study relating to walkability, there is still variation in what relationships exist between environmental factors and walking behaviour [8]. From the literature, there seems to be a difference on the restorative properties of a space based on the type of recreational activity.

A method for estimating trail traffic using recorded data was presented in [12] by extrapolating correlated features from a large dataset although it lacked in resolution for individuals and tracking source-destination pairs. It is important to note, however, the recent studies on evaluating the built environment related to comfort or attractiveness of a route are surveys with predefined questions (e.g., [6, 8]) and historically the majority of analysis methods were either surveys or interviews (see the review on Landscape and Urban Planning pre-2008 [15]). The importance of this is that the predefined aspect may not be interpreted in relation to sun conditions. For example, Deelen et al. (2019) found a positive association with poor lighting and attractiveness of the environment [6]. While the researchers explain a possible reason, the question (which the authors state is based on [8]) was asking about street lighting. Likewise, these surveys lack resolution in the time of a recreational run as the reported categories are: before 9am, after 9am, afternoons, evenings. Depending on the time of year and how much *before 9am* it is, the condition of lighting can drastically change.

In Renswouw et al. (2019), the difference in GPS trails of people running showed that some areas in which people run during the day are avoided at night [21]. Hot climates (e.g., Singapore) may have significant differences in recreational running patterns than more moderate areas as suggested by Balaban and Tunçer (2017) – runners avoid the time of day the sun is strongest [2]. With this assumption, the increase in shade along a running path would be beneficial to facilitating this activity during these intense sunlight hours. Comparatively, a cooler climate may want to promote more direct sunlight exposure, while still avoiding glare. Studying attributes of parks that lead to physical activity, Brown et al. (2014) used a method of public participation geographic information system (PPGIS) technology to overcome shortcomings with survey data, but did not present an analysis related to sunlight [5].

With the known benefits of sun exposure, the impact of radiant heating and various climates, and physiological response to glare, the evaluation metrics of a path in this paper are well established for their merit, although have not been directly studied for jogging.

2.2 Path Planning

Computing recreational routes (i.e., walking, jogging, running, cycling) in a city has been demonstrated most often in

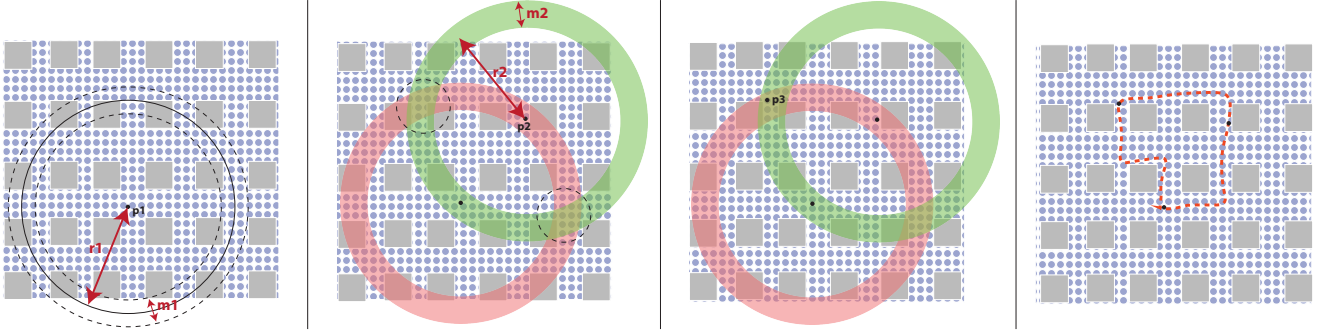


Figure 2: Visualization of waypoint selection process (from left to right) of a single given starting position. The process in steps 2,3,4 is applied for every node in the initial ring of step 1.

the field of computer graphics [10, 13, 22, 20]. However often in these works, particularly in the domain of human computer interaction, the emphasis is on the real-time generation of a single path for a user (e.g., on a phone app). The input data is largely based on maps (e.g., openstreetmap) and has little access to the three dimensional aspect of the built environment. The weights and environmental factors considered for the *goodness* of a route are often points of interest (POI) and influence the way in which the path is created, for example, in tourist routes [3, 4]. Sabine (2018) uses a method to query a source-destination shortest path with visible regions of interest without forcing the path to route through the POI directly [19]. Birsak et al. (2014) generates a tourist brochure using route planning between all pairs of POIs [3]. Their work intentionally reduces the graph iteratively as to reduce the number of overlapping routes.

A recommendation system for route planning of a closed cycle was shown in [13]. The framework also included numerous metrics to integrate in the final rankings (i.e., length, uniqueness, shape, lighting, elevation, pedestrian friendliness, turns, nature, history). In the case of lighting, the streets containing street lights were used and this metric was ignored during daytime. Similarly, Willamowski et al. (2019) focused on elevation as a driving input for the route planning of runners, emphasizing the generation of a new path using user preferences during construction (computationally, not physically) of the path compared to a final filtering step [22].

Waypoints and action descriptions (e.g., goals or tasks to complete) are often used in agent-based simulations to drive the agent in a particular direction. In optimizing sign placement for wayfinding, Huang et al. (2017) used a graph with paired source-destination points such as the bus stop to a POI, and POIs to each other [11]. The nodes in the graph are based on road segments, a common approach of urban path planning and analysis. While much of the work in route planning for multiple POIs involves the computation of an *attractive* path, this goal does not align well with the evaluation of an environment. For example, Birsak et al. (2017) plans routes that may not be the shortest, but rather are more interesting by passing multiple POIs [4]. In the design of an urban environment however, if a large number of paths generated have

mostly uninteresting routes, it is the design itself that can be modified. Mathew et al. (2018) generates an environment to create a desired crowd simulation effect [14]. This is similar to the generative design approach when the optimization metrics are evaluative scores on the environment.

While not integrating path planning, to analyze the impact of shadows in the urban environment, Miranda et al. (2018) introduced shadow accrual maps as a fast method for visualizing overlapping shadows in a city for city planners and architects [17]. This evaluation method integrated the three dimensional aspect of the buildings as it was similarly geared towards spatial evaluation for designers and planners.

Differences in both the initial data and goal of the path planning present a gap in the current state of the art relating to jogging route generation. As the evaluation of a 3D model for design purposes is evaluating possibilities of multiple people, the distance between locations is fully computed. Similarly, evaluating the impact of a building design on the human experience is done with the intention of understanding how changing that building will also impact people. Therefore the dimensionality is critical, rather than the network by itself.

3 METHODOLOGY

To provide an overall evaluation of diversity among loop-based jogging routes in the environment, three main steps are taken. First, the traversable surfaces of the environment are discretized into a grid-based dense graph in which the nearest neighbors are connected (eight neighbors) with an edge cost equal to the euclidean distance using the method in [18]. For the mass model type used in this paper and output by CityEngine, these surfaces are the sidewalks and crosswalks. Once this graph is created, Dijkstra’s All-Pairs-Shortest-Path (ASPS) algorithm is run to get the distance from all nodes to each other. The next step is to find a sequence of nodes that satisfy a user-specified distance a jog should be such that the sum of edge costs in the sequence are within some threshold of the total distance. This distance is input as a parameter composed of speed of the jog, desired length of time, and a percentage of tolerance to that length of time. The loop-like jogging routes available from a given point (e.g., a home in the environment) is calculated by finding the nodes within

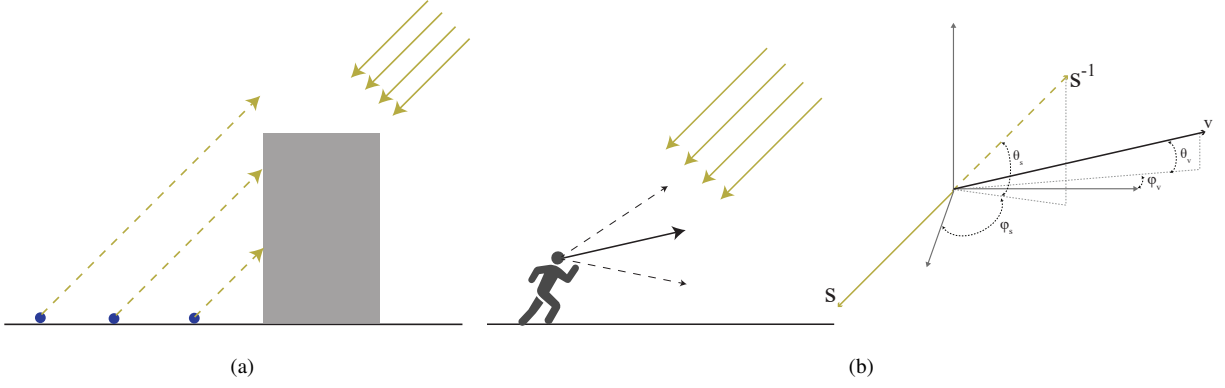


Figure 3: (3a): Illustration of the inverse sun vector being cast from a node to determine if it is within a shadow or not. (3b): Calculation method for determining if a direction a person is moving would be subject to glare from the sun.

one third the total distance and constructing a boolean vector for each of those valid nodes. This vector is then multiplied against each row of the distance matrix in which there is a corresponding true element, providing a set of tertiary nodes for the sequence.

Once the routes are created, the two evaluation metrics are calculated for a series of time periods defined by a user. First, the nodes that belong to the set of jogging routes are checked to be in a direct view of the sun or not (shadow). This is done by casting a ray in the opposite direction of the sun vector, starting at the node location with an offset to a human eye level (essentially a *shadow ray*). If the ray intersects any geometry (e.g., buildings) within the scene, it must be in shade. The next step is to take the set of edges, which are ordered node pairs, where the second node is not in shade, and compare the vector from the first to second node with the Sun vector. As there are varying amounts of discomfort for glare based on the location within the field of view, the angle threshold between these two vectors is determined by a parameter entered by the user.

3.1 Paths Computation

Fundamentally there are an infinite number of ways a path can be created in a given length. For an urban environment discretized to a 1 or 2 meter grid, while not infinite, the number of possible paths of a jogging length (e.g., 1000 meters) is impractical to calculate. More importantly, these path options are impractical to use in a meaningful way. For example, from a given location, walking between two nodes 1000 times would qualify. Techniques for the generation of paths often use a cost structure and find the minimal cost between two points. Rather than creating rules such as adding costs to edges that have been traversed, this paper uses a method of selecting sets of nodes that are inherently at appropriate distances. The general approach to finding the waypoints of the cycle is similar to the 2-via-routes ring method in [10] with a modified ring distance and precomputed APSP distance matrix.

A graph representation of the environment provides a discretized value of distances between locations that can be

considered for a path. For evaluation based on pedestrian movement, these locations are within the bounds of sidewalks and crosswalks. From this graph, an All Pairs Shortest Path (APSP) algorithm, in this case Dijkstra, to generate a distance and predecessor matrix. The distance matrix is an $n \times n$ matrix corresponding to the distances between each node such that a row i contains the distances of v_i from $v_1 \dots v_n$ and where the distance value of v_i to v_i is 0. The predecessor matrix is structured the same as the distance matrix except that instead of the distance between the two given nodes, it is the node with the minimal cost to it.

The distance matrix can then be used to find the set of nodes in the graph that are within some distance threshold from it. In the case of jogging path calculations, a desired distance can be given directly, or as implemented in here, a combination of jogging speed and desired time. To aid in the user experience of the grasshopper plugin a pedestrian speed of kilometers per hour and the total time in minutes is entered. As the distance matrix is set in meters, the total distance calculation is $T_d = k * 16.6667$. While T_d can be used to find all nodes from the start location that are the desired distance away, it would not account for the pedestrians travel time back. Unlike a single source-destination pair, the start and end location is the same, and when considering the distance matrix that is calculated by the Dijkstra APSP, the value of the same node to itself is 0. To resolve the distance discrepancy the easiest solution is to divide T_d by half. While the directionality would be useful in this approach, the path itself would remain as a line and have little variation for the pedestrian. Instead, waypoints are created to divide the path into multiple segments.

In the case of a path created by three nodes n_1, n_2, n_3 the predecessor matrix will be used to calculate the path of $n_1 \rightarrow n_2$, $n_2 \rightarrow n_3$, and $n_3 \rightarrow n_1$. Therefore, a path can be constructed by selecting three evenly distanced nodes of the predecessor matrix. The exact distance is unlikely to exist and therefore a margin of error must be incorporated. This approach uses four variables; a radius from the start node r_1 , a margin threshold for variation of that radius m_1 , a secondary radius defined from the second node r_2 , and likewise a second margin m_2 .

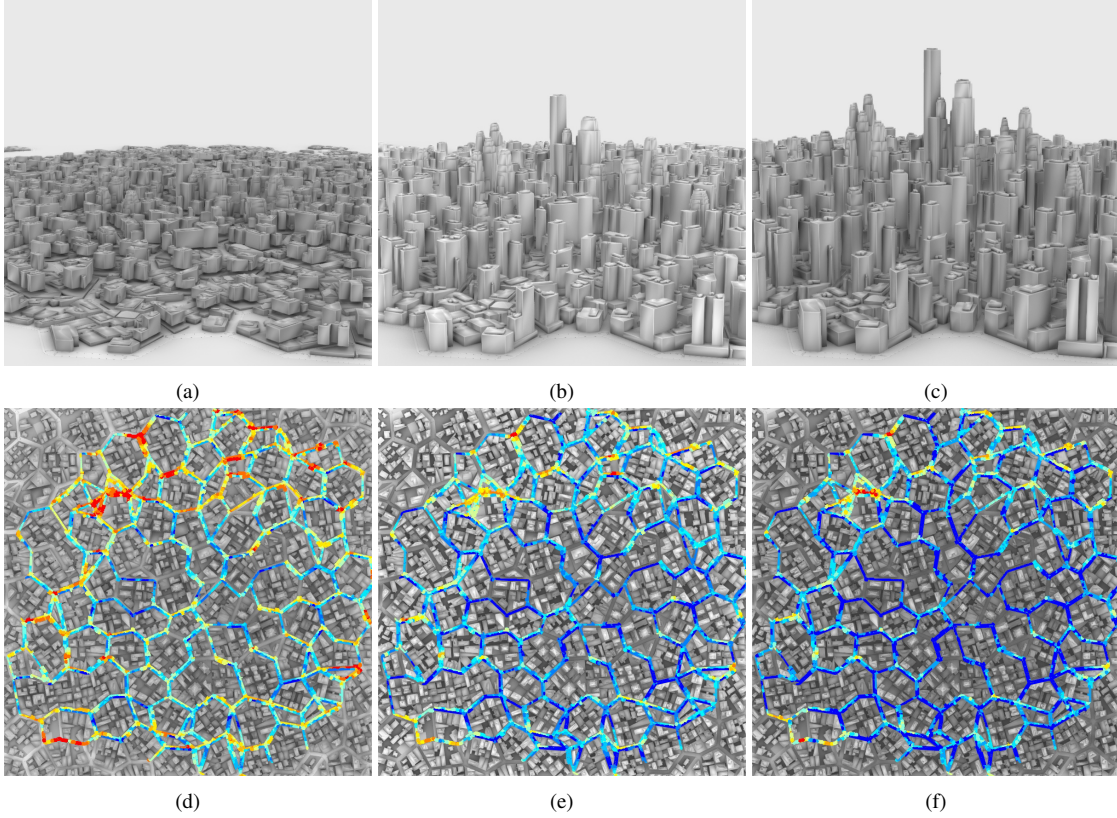


Figure 4: (4a-4c) perspective view of the generated city with increasing building heights from ESRI CityEngine. (4d-4f) A top view of the city (corresponding to the city directly above each image) with accumulative shadow scores calculated by the average number of times a node was in shadow or not over the designated sample period defined in the input parameters. The color is based on a low occurrence of times in the sun (blue) to high rate of occurrence (red).

In Figure 2, the process of finding the waypoints is demonstrated. First a node (referred to as point p_1) is selected as the start and end. The set of possible secondary nodes L are found by checking which nodes in the distance matrix are within the threshold, such that $\{v \in L : r_1 - (m_1/2) > D_{p_1}(v) > r_1 + (m_1/2)\}$, where $D_{p_1}(v)$ is the distance of a node v from p_1 . The next step is for each possible secondary point, (in Fig. 2, one example is shown labeled p_2 which falls within the red zone of p_1) find the final third node within the threshold distance. To be able to accurately retrieve a three node combination of a specified distance, the set of tertiary nodes is intersected with the secondary node set (dotted circles in Fig. 2).

The final step in Figure 2 is the dotted red line representing a calculated path that avoids obstacles such as buildings (as there are no nodes there), and generates an appropriately long path. The values of r_1 and r_2 are functionally independent. However in this work they are set to be equal as this creates evenly distanced waypoints such that $D_{p_1}(p_2), D_{p_2}(p_3), D_{p_3}(p_1) \approx T_d/3$ and a mostly circular route is formed within the city.

After generating all the paths, a random sample is taken for visualization and analysis in Rhino, assuming the desired

number of paths is less than the total calculated. As the values of m_1 and m_2 increase, the total number of paths increases as well. In rough terms, on a graph of $\approx 50,000$ nodes, $\approx 9,000$ paths are calculated in ≈ 10 seconds. From the sampled paths, any node used in a path is stored for analysis. The use of this path set has the additional benefit of reducing the calculations needed of the entire graph as only the nodes associated with a jogging path are used in the jogging path analysis.

3.2 Evaluation Metrics

The first evaluation metric is determining which nodes along the path are in shade. As this is independent of the path, all nodes can be calculated simultaneously without regard for order. A range of values for time of day are supplied, including the month, longitude, and latitude. For all the dates in range, the sum of the time a node is within shadow or not is stored. For determining the node is in a shadow or not for a given time, the inverse vector of the sun direction is cast as a ray from the node and checked for intersections with the models in the environment (Fig.3a).

The second evaluation metric is to determine if an edge along a path would be within a specified range such that the sun would cause discomfort from glare to a person. As illustrated

in Figure 3b, the view direction of a person is estimated by the vector of the ordered nodes making up an edge in a path. If both nodes of an edge are in shadow, the glare calculation can be skipped as the sun would not be visible even if the vectors are the same. A range determined by the user of the plugin for determining the cutoff for glare is used as the threshold t_θ, t_φ between the Sun and human view vectors. To make the calculation method understandable and transparent, the cartesian space-based vectors are converted into spherical coordinates. Similar with the shadow calculation, the comparison of view to the sun vector s is done with the inverse of the vector s^{-1} . The x,y,z coordinates are then converted to azimuth φ , elevation θ , and a radius (which is ignored for unit vectors of length 1). The sun is within the glare range if $abs(\theta_v - \theta_s) < t_\theta$ and $abs(\varphi_v - \varphi_s) < t_\varphi$.

To verify the sampling choice of 300 paths to be adequate for this preliminary analysis and demonstration, the evaluation metrics for shade and glare of two sets of separately generated paths are plotted as histograms in Figure 5. As evident in the image, the consistency between the two sets is strong.

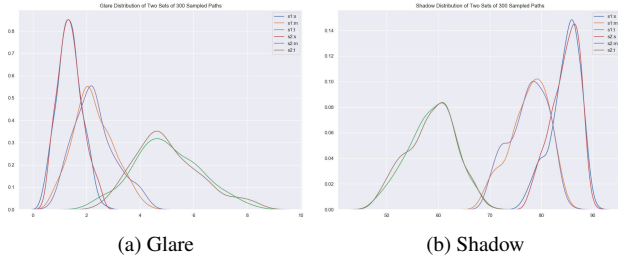


Figure 5: Histogram comparison between two sets of 300 paths sampled from the total possible with the shadow analysis.

4 RESULTS

With the aggregated jogging paths shown in Figure 1, the choice of certain sidewalks and streets over others can be seen in relation to the given starting point (where the blue and red parts of the paths meet). While it is clearly possible one would choose to use these side streets, the algorithms results are logical in that the most underutilized streets are perpendicular to the outgoing radial lines. To further illustrate the types of paths that are created, individual examples are given in Figure 6.

The example city used for this demonstration was generated in ESRI CityEngine [7]. A hexagonal grid was used with three variations of the same city in which the building height scale was modified (Fig. 4a-4c). The height is a multiplying factor from the original, rather than an overall addition of floors. This demonstrates the impact of relatively similar sized buildings (Fig. 4a) to one in which the city center has skyscrapers (Fig. 4c).

Based on the peak times and month for urban trail utilization in [12], the following demonstration of results utilize a time of day ranged from 06:00 - 18:00 during August 1st, 2019. The longitude and latitude were set as 44.725° and -74.919°

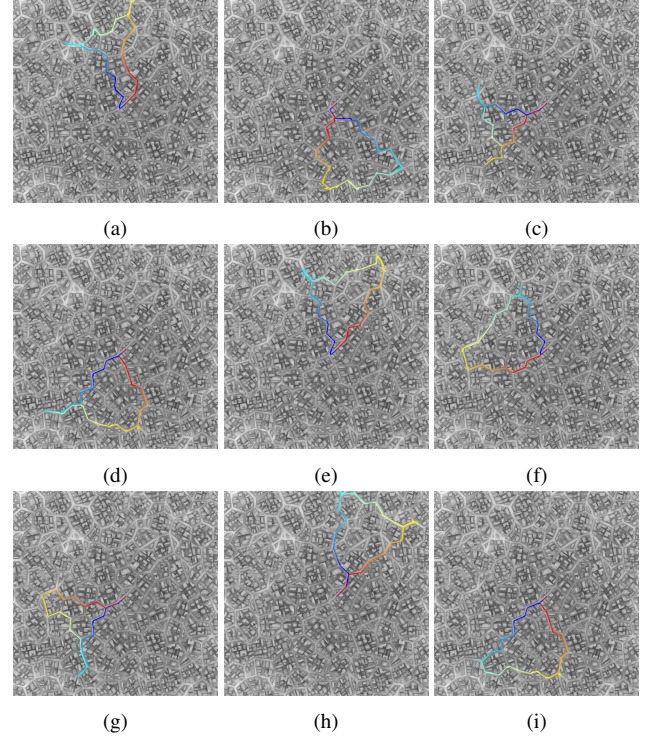


Figure 6: Sample Jogging Paths. Each path is colored by the distance from the start, where blue is the beginning and red is the end.

respectively. The impact of sun glare to a human varies by angle, however a threshold of 50° is used for the results here. For this example, the jogging speed was defined as 7km/hr (a light jog) and the city evaluation was performed for 15 minute jogs (7km/hr = 116.7 m/min), at 1750 meters. Each margin as described in Sec. 3 were set to 15%, resulting in a jogging path range between $\approx 1485 - 2000$ meters.

The visualization of the shadow parameters is shown in Figures 4d-4f. In this visualization it is easy for a designer to see the locations in which sun will often be on a node or not. Depending on the design intention, this could be an opportunity to add shade (naturally e.g., with trees) or emphasis for the jogging route. The shorter building heights in Fig. 4a correspond to the shadow scores in Fig. 4d. Comparing Fig. 4d to Fig. 4f, the individual normalized values (i.e., the normalization in the colors are not impacted by the design alternatives) demonstrate how the larger buildings within the center provide very little sunlight compared to more open areas near the outer limits of the jogging paths.

One method of understanding the relationship between the urban design options is through the histogram plots. As seen in Figure 7, the difference in percentage of the jogging paths in shadow are significantly different between the three building height models. For the small building heights (colored in green), the range of path percentage in shadow is from $\approx 50\%$ to $\approx 70\%$ while the tallest building variant (colored in blue) ranges from $\approx 70\%$ to $\approx 90\%$. Just as noticeable is the re-

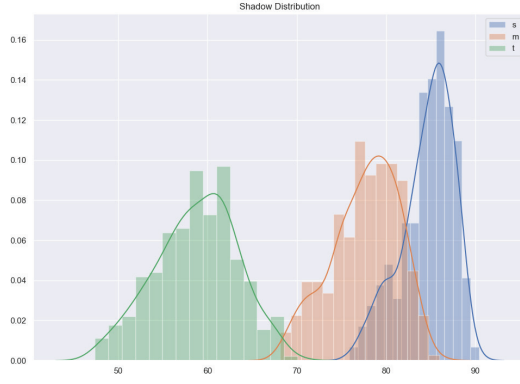


Figure 7: Histogram with kernel density estimation for the percentage of nodes of each jogging path that are in shadow for the three building heights. The colors correspond as follows: Fig. 4a-green, Fig. 4b-orange, Fig. 4c-blue.



Figure 8: Histogram with kernel density estimation for the percentage of edges in each jogging path that are within the glare range for the three building heights.

duced distribution of bins between the two options. If consistency along a path is desired, the more concentrated values of option 4c may be considered better.

The evaluation scores based on the percentage of the paths that contain sun glare are similarly proportioned with slightly more overlap, seen in Figure 8. As may be expected, the tall building city variant has very little of the path in the glare range. While demonstrated in this paper is the difference in building heights, alternative design options related to glare could include the orientation of the street network. As the angle of glare is dependent on the direction of the path, an environment with hills and a surface terrain that is not flat would likely have a larger impact in the variation of glare (which the current framework supports).

To illustrate the importance of the utilization of sidewalks and crosswalks for the evaluation, a comparison between a graph that is generated within a 1 meter road and the sidewalks was done. In both cases, 300 samples were taken of the calculated jogging paths with the same approximate source-destination

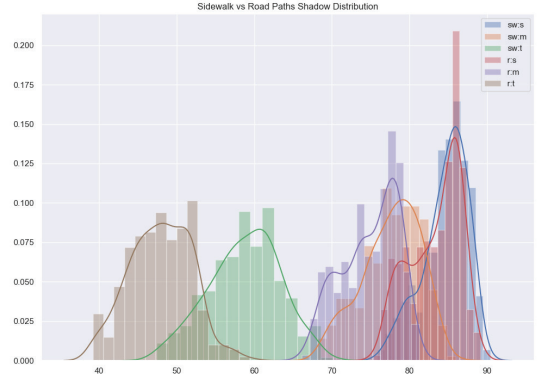


Figure 9: Comparison of histograms for the percentage of all paths that are in shadow when using the road center vs the sidewalk.

location. In Figure 9 the two sets of results are shown for the shadow percentage along paths.

It is easy to see why the large difference in road vs sidewalk evaluations exist when considering Figure 3a as the points closest to the building will have a larger angle in which the sun vector must be to provide a line of sight. When the center of the road is used, the shadow cast by a building is less likely to impact it.

5 DISCUSSION

This paper presented a method for generating paths within a user-specified distance that start and end in the same location (i.e., jogging paths). Two methods for evaluating the paths are presented; the percentage of a path in shadow and percent facing the sun within a glare range. An insightful result from the analysis shows the large difference in the shadow-based analysis metric when using paths generated from a road surface (as is the case with road centerlines) compared to on the sidewalk. Further analysis and comparisons using this generation and evaluation technique will include terrain with more variation such as hills as the angle of view will more drastically change during human mobility.

In the urban design discourse, walkability is a frequently used term and method of analyzing the built environment, and subsequently, as a means to evaluate the difference between design options. With the broad scope in definition and metrics being used, this paper introduces two more: metrics that are related to the directionality of movement and not just distance between a single source and different destination. This is combined with element of time in which both the position of the sun and direction of the persons movement are incorporated. When sampling the sun position at 60 minute intervals, a route that takes 15 minutes to complete has no change in the sun vector. The evaluations provided by the framework presented here would then be logical only within that sampling period. Future work will investigate the impact of using the change in sun vectors over the duration of the jogging path.

While the shade and glare metrics can be applied to any path, the method for jogging paths is logical to the assumption that

a person starts and ends at their own home. As seen in the literature, often people will intentionally jog around a designated area such as a park, in which this current method does not account for. The motivation here is that an environment should not limit the potential for numerous variants of 15 minute jogging routes to beginning within a park.

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